

RAHUL REDDY YERRAM

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Software Engineer with 3+ years of experience in full-stack development, cloud-native microservices, and real-time streaming pipelines, skilled in Java, Python, TypeScript, React.js, Spring Boot, GCP and Docker across enterprise and startup environments.

EDUCATION

Master of Science in Data Science | Monash University

Melbourne, VIC | July 2024 - July 2026

GPA: 3.25 / 4.0 | Coursework: Python Programming, Database Systems, Big Data Processing, Computer Architecture

Bachelor of Engineering – Mechanical Engineering | B.M.S College of Engineering

Bangalore, India | July 2018 - August 2022

GPA: 9.0 / 10 | Electives: Machine Learning, AI, Robot Process Automation, Python Programming, Web Technologies

SKILLS

- **Languages & Frameworks:** Python, Java, TypeScript, JavaScript, React.js, Vue.js, Spring Boot, FastAPI, Node.js, HTML, CSS
- **Cloud & DevOps:** GCP, AWS, Docker, CI/CD, Git, Linux CLI
- **Databases & Tools:** PostgreSQL, MySQL, Cloud Spanner, REST APIs, Jira, Postman
- **Other:** Apache Kafka, PySpark, OpenCV, TensorFlow, NLP, ML, AI, ETL, Power BI

WORK EXPERIENCE

Software Engineer | ADPMN

Melbourne, VIC | November 2025 – February 2026

- Built and maintained key frontend modules for an E-Commerce Platform using **React.js**, **HTML**, **CSS**, and **JavaScript**, supporting dynamic product browsing, search, and cart management.
- Cut order processing time by **30%** by building and deploying an Order Management Microservice using **Java Spring Boot** on **GCP Cloud Run**.
- Reduced release cycle time by **40%** by setting up **CI/CD** pipelines on **Google Cloud Build** with automated triggers, eliminating manual deployment errors.
- Set up isolated Dev, QA, and production on **GCP Cloud Run** from scratch, managing infrastructure including **Google Secret Manager**, **Cloud Spanner**, **GCP Storage Buckets**, and **Artifact Registry**.

Software Engineer | MCAV – Monash Connected Autonomous Vehicles

Melbourne, VIC | March 2025 – January 2026

- Built a real-time autonomous vehicle pipeline for **IGVC**, Michigan using **OpenCV**, **ROS2**, and **LiDAR** sensor fusion on **Ubuntu**, reducing lane detection failures by **20%** and hardware testing cycles by **30%**.

Software Development Engineer I | Dassault Systèmes

Bangalore, India | March 2023 – June 2024

- Delivered upload progress tracking, 3D model viewer, and asset annotations for xMedia on 3DEXPERIENCE by developing and integrating **Java REST APIs** with **TypeScript** and **JavaScript** frontend under Agile sprints.
- Improved platform reliability to 91% code coverage by building **Jasmine** unit test suites and expanded global reach by delivering **NLS** (Native Language Support) for full multi-language localisation.
- Automated manual metadata classification by building a **CNN**-based Auto Tag Suggestions POC using **TensorFlow.js** and **COCO** dataset, achieving 92% accuracy.

Software Engineer Intern | Dassault Systèmes

Bangalore, India | July 2022 – March 2023

- Built a standalone **JavaScript** application from scratch to parse **EXIF** metadata from **PNG**, **JPEG**, and **MP4** formats without any existing library support, resolving a critical cross-functional workflow gap while fixing 1,780+ bugs with zero critical defects in production.

Machine Learning Engineer Intern | Toyota Kirloskar Motor Pvt. Ltd.

Bangalore, India | April 2022 – July 2022

- Engineered **Python** automation scripts using **Selenium** and **BeautifulSoup** to scrape and extract vehicle diagnostic data from Field Technical Reports, eliminating manual workflows and boosting team productivity by **35%**.
- Built and deployed a fuel injector clogging detection software using **Deep Learning** to automatically classify clogging ranks, improving QA inspection accuracy by **20%**.
- Developed a web scraping pipeline feeding a **Power BI** dashboard for automated root cause analysis across customer fault data, improving QA response efficiency by **30%**.

PROJECTS

- Transport-Aware Social Discovery Platform:** Built a full-stack application using **Vue.js**, **FastAPI**, and **PostgreSQL** integrating Eventfinda, OpenStreetMap, and PTV GTFS APIs for real-time event discovery and transport planning and **CI/CD** via **GitHub Actions** across **Vercel** and **AWS EC2**.
- Real-Time Energy Prediction Pipeline:** Built a distributed streaming pipeline using **Apache Kafka** and **PySpark** Structured Streaming with a 5-second watermark for fault-tolerant data handling, delivering live building-level energy aggregates across 16 sites at 2Hz refresh rate within a **Dockerised** environment.
- Global Wildfire Emissions Analysis:** Built scalable **ETL** pipelines in **R** and **Python** processing 4 datasets (1.5M+ records) with **IQR**-based anomaly detection across 120.5M GlobFire records, feeding an interactive **D3.js** dashboard visualising 13 years of global emission trends.

ACHIEVEMENTS

People's Choice Award – IDEATHON 2023 (Global), Dassault Systèmes - first AI integration in xMedia using TensorFlow.js